**East West University**

**Department of Computer Science and Engineering**

**Summer 2022**

**Lab Assignment Submission Report-01**

**ASSIGNMENT TITLE:** Introduction to Oracle Database XE and SQLPlus Tool for writing SQL queries

**Course Code:** CSE302

**Course Title:** Database Systems

**Section:** 02

**SUBMITTED BY**

**Student Name:** Jannatul Ferdous Salma

**Student ID:** 2020-2-60-005

**SUBMITTED TO**

Mohammad Rezwanul Huq, PhD

Associate Professor,

Department of Computer Science and Engineering

**DATE OF SUBMISSION**

**17 JUNE 2022**

# Screenshots

## Lab Task # 01 (Creating a table):

1.(a). Write SQL statement to create a table ‘instructor_2020260005’ which has 4 attributes –

i) id (number)

ii) name (text)

iii) dept_name (text)

iv) salary (number)

```
SQL> CREATE TABLE instructor_2020260005(
  2       id NUMBER,
  3       name VARCHAR2(25),
  4       dept_name VARCHAR2(25),
  5       salary NUMBER,
  6       PRIMARY KEY(id)
  7   );

Table created.
```

Figure:01

1.(b). Write SQL statement to create a table ‘course_2020260005’ which has 4 attributes –

i) course_id (text)

ii) title (text)

iii) dept_name (text)

iv) credits (number)

```
SQL> CREATE TABLE course_2020260005(
  2      course_id VARCHAR2(25),
  3      title VARCHAR2(70),
  4      dept_name VARCHAR2(70),
  5      credits NUMBER,
  6      PRIMARY KEY(course_id)
  7  );

Table created.
```

Figure:02

## Lab Task # 02 (Inserting data into a table):

2.(a). Write SQL statements to insert the following records into ‘instructor_2020260005’ table:

| ID | name | dept_name | salary |
|---|---|---|---|
| 10101 | Srinivasan | Comp. Sci. | 65000 |
| 12121 | Wu | Finance | 90000 |
| 15151 | Mozart | Music | 40000 |
| 22222 | Einstein | Physics | 95000 |
| 32343 | El Said | History | 60000 |
| 33456 | Gold | Physics | 87000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 58583 | Califieri | History | 62000 |
| 76543 | Singh | Finance | 80000 |
| 76766 | Crick | Biology | 72000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 98345 | Kim | Elec. Eng. | 80000 |

```
SQL> INSERT INTO instructor_2020260005 VALUES(10101,'Srinivasan','Comp.Sci',65000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(12121,'Wu','Finance',90000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(15151,'Mozart','Music',40000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(22222,'Einstein','Physics',95000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(32343,'El Said','History',60000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(33456,'Gold','Physics',87000);

1 row created.
```

Figure:03

```
SQL> INSERT INTO instructor_2020260005 VALUES(45565,'Katz','Comp.Sci',75000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(58583,'Califieri','History',62000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(76543,'Singh','Finance',80000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(76766,'Crick','Biology',72000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(83821,'Brandit','Comp.Sci',92000);

1 row created.

SQL>
SQL> INSERT INTO instructor_2020260005 VALUES(98345,'Kim','Elec.Eng.',80000);

1 row created.
```

Figure:04

2 (b). Write SQL statements to insert the following records into ‘course_2020260005’ table:

| *course_id* | *title* | *dept_name* | *credits* |
|---|---|---|---|
| BIO-101 | Intro. to Biology | Biology | 4 |
| BIO-301 | Genetics | Biology | 4 |
| BIO-399 | Computational Biology | Biology | 3 |
| CS-101 | Intro. to Computer Science | Comp. Sci. | 4 |
| CS-190 | Game Design | Comp. Sci. | 4 |
| CS-315 | Robotics | Comp. Sci. | 3 |
| CS-319 | Image Processing | Comp. Sci. | 3 |
| CS-347 | Database System Concepts | Comp. Sci. | 3 |
| EE-181 | Intro. to Digital Systems | Elec. Eng. | 3 |
| FIN-201 | Investment Banking | Finance | 3 |
| HIS-351 | World History | History | 3 |
| MU-199 | Music Video Production | Music | 3 |
| PHY-101 | Physical Principles | Physics | 4 |

```
SQL> INSERT INTO course_2020260005 VALUES('BIO-101','Intro. to Biology','Biology',4);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('BIO-301','Genetics','Biology',4);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('BIO-399','Computational Biology','Biology',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('CS-101','Intro. to Computer Science','Comp. Sci',4);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('CS-190','Game Design','Comp. Sci',4);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('CS-315','Robotics','Comp. Sci',3);

1 row created.
```

Figure:05

```
SQL> INSERT INTO course_2020260005 VALUES('CS-319','Image Processing','Comp. Sci',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('CS-347','Database System Concepts','Comp. Sci',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('EE-181','Intro. to Digital System','Elec. Eng.',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('FIN-201','Invesment Banking','Finance.',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('HIS-351','World History','History.',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('MU-199','Music Video Presentation','Music.',3);

1 row created.

SQL>
SQL> INSERT INTO course_2020260005 VALUES('PHY-101','Physical Principles','Physics.',4);

1 row created.
```

Figure:06

## Lab Task # 03 (Writing Queries):

i. Show instructor name only.

```
SQL>  SELECT name FROM instructor_2020260005;

NAME
-------------------------
Srinivasan
Wu
Mozart
Einstein
El Said
Gold
Katz
Califieri
Singh
Crick
Brandit

NAME
-------------------------
Kim

12 rows selected.
```

Figure:07

ii. Show course id and title only.

```
SQL> SELECT course_id, title FROM course_2020260005;

COURSE_ID
-------------------------
TITLE
----------------------------------------------------------------------
BIO-101
Intro. to Biology

BIO-301
Genetics

BIO-399
Computational Biology


COURSE_ID
-------------------------
TITLE
----------------------------------------------------------------------
CS-101
Intro. to Computer Science

CS-190
Game Design

CS-315
Robotics
```

Figure:08

```
COURSE_ID
-------------------------
TITLE
----------------------------------------------------------------------
CS-319
Image Processing

CS-347
Database System Concepts

EE-181
Intro. to Digital System


COURSE_ID
-------------------------
TITLE
----------------------------------------------------------------------
FIN-201
Invesment Banking

HIS-351
World History

MU-199
Music Video Presentation


COURSE_ID
-------------------------
TITLE
----------------------------------------------------------------------
PHY-101
Physical Principles


13 rows selected.
```

Figure:09

iii. Find instructor name and department of the instructor with id = 22222.

```
SQL> SELECT name,dept_name FROM instructor_2020260005
  2      WHERE id=22222;

NAME                      DEPT_NAME
------------------------- -------------------------
Einstein                  Physics
```

Figure:10

iv. Find course title and credits of the courses offered by ‘Comp. Sci.’ department.

```
SQL> SELECT title,credits FROM course_2020260005
  2      WHERE dept_name='Comp. Sci';

TITLE
--------------------------------------------------------------------
   CREDITS
----------
Intro. to Computer Science
         4

Game Design
         4

Robotics
         3


TITLE
--------------------------------------------------------------------
   CREDITS
----------
Image Processing
         3

Database System Concepts
         3
```

Figure:11

v. Find name and department of instructors who have a salary more than 70000.

```
SQL>  SELECT name,dept_name FROM instructor_2020260005
  2      WHERE salary>70000;

NAME                      DEPT_NAME
------------------------- -------------------------
Wu                        Finance
Einstein                  Physics
Gold                      Physics
Katz                      Comp.Sci
Singh                     Finance
Crick                     Biology
Brandit                   Comp.Sci
Kim                       Elec.Eng.

8 rows selected.
```

Figure:12

vi. Find course title of the courses which are not less than 4 credits.

```
SQL>  SELECT title FROM course_2020260005
  2      WHERE credits>=4;

TITLE
--------------------------------------------------------------------
Intro. to Biology
Genetics
Intro. to Computer Science
Game Design
Physical Principles
```

Figure:13

vii. Find name and department of instructors who have a salary in between 80000 and 100000 (bounds are inclusive).

```
SQL> SELECT name,dept_name FROM instructor_2020260005
  2      WHERE salary BETWEEN 80000 AND 100000;

NAME                      DEPT_NAME
------------------------- -------------------------
Wu                        Finance
Einstein                  Physics
Gold                      Physics
Singh                     Finance
Brandit                   Comp.Sci
Kim                       Elec.Eng.

6 rows selected.
```

Figure:14

viii. Find course title and credits of the course not offered by 'Comp.Sci' department

```
SQL> SELECT title,credits FROM course_2020260005
  2      WHERE dept_name != 'Comp. Sci';

TITLE                        CREDITS
------------------------- ---------
Intro. to Biology                 4
Genetics                          4
Computational Biology             3
Intro. to Digital System          3
Invesment Banking                 3
World History                     3
Music Video Presentation          3
Physical Principles               4

8 rows selected.
```

Figure:15

ix. Display all records of instructor table.

```
SQL> SELECT *
  2      FROM instructor_2020260005;

        ID NAME                      DEPT_NAME                     SALARY
---------- ------------------------- ------------------------- ----------
     10101 Srinivasan                Comp.Sci                       65000
     12121 Wu                        Finance                        90000
     15151 Mozart                    Music                          40000
     22222 Einstein                  Physics                        95000
     32343 El Said                   History                        60000
     33456 Gold                      Physics                        87000
     45565 Katz                      Comp.Sci                       75000
     58583 Califieri                 History                        62000
     76543 Singh                     Finance                        80000
     76766 Crick                     Biology                        72000
     83821 Brandit                   Comp.Sci                       92000

        ID NAME                      DEPT_NAME                     SALARY
---------- ------------------------- ------------------------- ----------
     98345 Kim                       Elec.Eng.                      80000

12 rows selected.
```

Figure:16

x. Find all courses (display all columns) which are offered by ‘Biology’ department and credits is not 4.

```
SQL> SELECT *
  2      FROM course_2020260005
  3      WHERE dept_name = 'Biology' AND credits != 4;

COURSE_ID
-------------------------
TITLE
--------------------------------------------------------------------------
DEPT_NAME
--------------------------------------------------------------------------
   CREDITS
----------
BIO-399
Computational Biology
Biology
         3
```

Figure:17

**Learning Outcomes:** The following is what we have learned in this lab.

**1)** How to install Apex Oracle database Software.

**2)** How to use Oracle database and SQLPLUS command tool.

**3)** How write all SQL statements in notepad first and save them with the .sql extension then use the command Prompt.

**4)** We have learned to make a common user and grant privileges.

**5)** We also learned how to create a table, insert data into a table, and some queries.

**6)** Then finally, we have completed the lab work.